

Tangail Saree E-Commerce Website

Product & Technical Requirements Document

V1 — MVP / Initial Launch

1. Business Overview

The business will sell Tangail sarees online using a Facebook page for marketing and an e-commerce website for product presentation, ordering, and order management. The initial business model avoids holding large inventory because sarees can be sourced from a trusted friend/factory when an order is received.

2. Business Goals

- Build a simple, trustworthy online presence for Tangail sarees.
- Use Facebook/social media to attract customers and the website to convert visitors into orders.
- Allow customers to buy multiple saree designs and quantities in one order.
- Collect customer phone number and delivery address during checkout without requiring customer login.
- Provide an admin area for products, orders, customers, discounts, and order status.
- Keep the first version simple and inexpensive; add advanced features only after validating demand.

3. V1 Scope

Included in the first release:

- Customer-facing storefront
- Product catalog and product details
- Categories / filtering
- Shopping cart
- Multiple products and quantities per order
- Guest checkout
- Customer phone and delivery address collection
- Order creation and confirmation
- Simple discount/sale pricing
- Admin dashboard
- Product CRUD
- Order management
- Simple order status tracking
- Customer/order history
- Basic order status history

Not required initially:

- Customer account/login/OTP
- Live courier GPS tracking

- Courier API integration
- Complex coupon engine
- Wishlist
- Reviews and ratings
- Advanced analytics
- Large warehouse/inventory management system
- Mobile app

4. Customer Journey

1. Customer sees a saree on Facebook/social media.
2. Customer opens the website/product page.
3. Customer selects quantity and adds the item to cart.
4. Customer can add other sarees to the same cart.
5. Customer opens checkout.
6. Customer provides name, phone number, and delivery address.
7. System calculates item totals, discount, delivery charge if applicable, and final total.
8. Customer places the order.
9. Admin receives the order and confirms it.
10. Order progresses through simple statuses until delivery.

5. Main Website Pages

Home: Brand introduction, featured sarees, new arrivals, sale items, categories, trust information, and Facebook/social links.

Shop / Products: Product grid with category/filter/sort options.

Product Details: Images, product name, product code, description, saree details, price, sale price, availability, quantity selector, and Add to Cart.

Cart: All selected products, quantities, unit prices, discounts, subtotal, delivery charge, and total.

Checkout: Guest checkout form with name, phone, address, district/area, optional postal code, order summary, and place-order action.

Order Confirmation: Order number, summary, total, delivery information, and basic next steps.

Contact / About: Business information, contact details, social links, and brand story.

6. Admin Dashboard

- Dashboard overview: pending orders, confirmed orders, sales summary, and recent orders.

- Products: create, edit, delete/archive, upload images, manage price, sale price, category, product code, and availability.
- Orders: view order details, customer information, items, quantities, pricing, discount, delivery information, and current status.
- Order status: Pending → Confirmed → Preparing → Shipped → Delivered.
- Additional statuses: Cancelled and Returned.
- Order status history: save each status change with timestamp and optional note.
- Customers: view customer contact information and previous orders.
- Discounts: support simple product sale pricing and selected order-level discounts.
- Admin authentication: required for dashboard access.

7. Order & Cart Model

A cart is temporary. An order is permanent. One order must support many products and different quantities.

Product	Quantity	Unit Price	Discount	Subtotal
Tangail Saree A	2	₹2,500	₹200	₹4,800
Tangail Saree B	1	₹3,000	₹0	₹3,000
Tangail Saree C	3	₹1,800	₹300	₹5,100

Important: OrderItem should store the actual price/discount used when the order was placed. Changing a product's future price must not change the price of an old order.

8. Customer Data & Guest Checkout

V1 will not require customer accounts. The customer provides information at checkout. The system can associate repeat orders using the phone number.

- Customer: id, name, phone, createdAt, updatedAt.
- CustomerAddress: id, customerId, address, district, area, postalCode, isDefault.
- Order should store a delivery/customer snapshot so historical orders remain correct even if the customer later changes their address.

Do not add password/OTP authentication unless it becomes necessary after launch.

9. Product Data

- id
- name
- productCode / SKU
- description
- category
- regularPrice
- salePrice (optional)
- saleStartAt / saleEndAt (optional)
- images
- availability
- createdAt / updatedAt

10. Discount Strategy

Keep discounts simple in V1:

- Product-level sale price: regular price + sale price.
- Optional order-level discount, such as a fixed amount above a minimum order value.
- Future options: coupon codes, first-order coupon, bundle discounts, and flash sales.

The final order must save the applied discount and final unit price. Avoid misleading inflated regular prices.

11. Order Status

Status	Meaning	Typical Action
Pending	New order received	Admin reviews order
Confirmed	Order accepted	Source/prepare saree
Preparing	Order is being prepared	Package item
Shipped	Handed to courier	Add courier/tracking info if available
Delivered	Customer received order	Complete order
Cancelled	Order cancelled	Record reason if known
Returned	Order returned	Record return information

12. Suggested Database Structure

- Product
- Category
- ProductImage
- Customer
- CustomerAddress
- Order
- OrderItem
- OrderStatusHistory
- AdminUser
- Discount / Promotion (keep simple in V1)

Relationship example: Customer 1→many Orders; Order 1→many OrderItems; Product 1→many OrderItems; Order 1→many OrderStatusHistory; Customer 1→many CustomerAddresses.

13. Recommended Technology Stack

Area	Choice
Frontend	Next.js + TypeScript
UI	Tailwind CSS; shadcn/ui optional
Forms	React Hook Form + Zod
Backend	Next.js Route Handlers / Server Actions for V1
Database	PostgreSQL
ORM	Prisma
Authentication	Simple admin authentication; no customer authentication initially
Images	Object storage such as S3-compatible storage if

	needed
Deployment	Vercel or another suitable managed deployment platform
Version Control	Git + GitHub

Avoid Redux or a separate Express backend in the first version unless the application actually needs them.

14. Suggested Project Architecture

- app/ — pages, layouts, route handlers
- components/ — reusable UI components
- features/ — product, cart, checkout, orders, customers, admin modules
- lib/ — database, utilities, validation, storage helpers
- prisma/ — schema and migrations
- types/ — shared TypeScript types

Keep business logic separated from UI code. For complex operations, use a clear flow such as route/action → service/business logic → database access.

15. Business & Operational Flow

- Facebook/social media → Website
- Website → Customer order
- Admin → Confirm order
- Factory/source → Saree preparation
- Admin → Packaging
- Courier → Delivery
- Admin → Update order status

16. Inventory Approach for V1

Because sourcing is available through a friend/factory, the business does not need a complex warehouse system at launch. The website can initially use a simple availability field such as Available / Unavailable or an optional quantity field.

Before confirming an order, admin should verify that the requested saree is actually available. If demand grows, real inventory/reservation logic can be added later.

17. Delivery & Payment

- V1 can start with cash on delivery if that matches the business's courier arrangement.
- Store delivery charge separately from product subtotal.
- Keep courier name and tracking/reference number optional.
- Do not build live courier tracking initially.
- Online payment integration can be added after the basic sales flow is validated.

18. Brand & Marketing

The brand should use a simple, memorable Bengali/traditional name. A candidate discussed was “Tangail Tat Ghor / টাঙ্গাইল তট ঘর”. Before committing to packaging, signage, or major advertising, check Facebook/search presence, domain availability, and relevant trademark/brand conflicts.

Marketing channel: Facebook page first. The website should be linked from the page and product posts should deep-link customers to the relevant product page.

19. V1 Security & Reliability

- Protect all admin routes.
- Validate all customer input on the server, not only in the browser.
- Use Zod or equivalent validation.
- Never trust client-side price calculations; calculate final totals on the server.
- Prevent customers from changing product prices through request manipulation.
- Store order price snapshots.
- Use HTTPS and secure environment variables.
- Back up the database.

20. Development Roadmap

Phase	Work
Phase 1	Project setup, database schema, admin authentication
Phase 2	Product/category management and image handling
Phase 3	Home/shop/product detail pages
Phase 4	Cart and quantity handling
Phase 5	Guest checkout and order creation
Phase 6	Admin order management and status history
Phase 7	Discount/sale pricing
Phase 8	Testing, security checks, deployment
Phase 9	Facebook launch and customer acquisition

21. Future Features

- Customer accounts and OTP login
- Online payment gateway
- Courier API integration
- Automated SMS/WhatsApp notifications
- Coupon management
- Wishlist
- Reviews and ratings
- Advanced inventory management
- Sales and profit reports
- Facebook/Meta catalog integration
- Analytics and conversion tracking
- Multiple admin roles

22. MVP Success Criteria

- A customer can find a saree from Facebook and reach its product page.
- A customer can add multiple sarees and quantities to one cart.
- A customer can place an order without creating an account.
- Admin can see the complete order and customer information.
- Admin can update the order status.
- Historical orders retain their original product prices and customer delivery details.
- The business can operate the system without needing complex technical steps.

23. Key Principle

Build the smallest system that can reliably sell and manage saree orders. Validate the business first, then invest in advanced features, branding, automation, and inventory/payment integrations based on actual customer demand.